

### Course Information

Course Code: CS-08519

Credit Hours: 4 (3+1)

Batch: Fall-2025

Course Title: **Computer Networks**

Lecture Hours/Week: 3

Pre-Requisites: **None**

### c. Course Contents

| <b>Course Instructors</b>           |          | M. Nadeem Akhtar, M. Shahzad, M. Abdullah, M. Zaman Aslam, M. Faiez, Ms. Iqra Adnan, Ms. Sadia Hameed, Ms. Zoha Ahsan, Ms. Sana Saeed and Ms. Rahila Batool.                                                                                                                                                                                                                                                                                                           |
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| <b>Syllabus Designed By ( TAI )</b> |          | Mr. M. Nadeem Akhtar                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Week                                | Lectures | Topics / Sub-Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1                                   | 1 & 2    | <b>Data Communication Fundamentals</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     |          | What is Data Communication, Characteristics of Data Communication System, Components of Data Communication System, Data Representation, Data Flow modes<br><b>Computer Networks Overview</b><br>Brief Intro to Networks (definition of networks, nodes, links), Computer Network criteria (performance, reliability, security), Physical Structures of Networks (Based on type of connection, Based on Topology)                                                       |
| 2                                   | 1 & 2    | <b>Computer Networks Overview (Cont...)</b><br>Categories of Networks (LAN, MAN, WAN, PAN, BAN), Internet <b>Protocol Layering</b> : Layering Scenarios and Principles, Why layered architecture? Overview of OSI model.<br>TCP/IP Protocol Suit details, Addressing (Physical, logical, Port, Specific)                                                                                                                                                               |
| 3                                   | 1 & 2    | Transmission Media and Network Connecting Devices<br><b>Physical Layer</b><br>Data and Signals (Analog as well digital), Parameters of a sine wave (Peak amplitude, frequency/period, phase), Wavelength, Composite signals, Bandwidth, Bit rate, bit length<br>Transmission Impairments (Attenuation, Distortion, Noise),<br><b>Channel Capacity</b><br>Definition of channel and channel capacity, Effect of noise on channel capacity, Signal to noise ratio (SNR), |

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| 4  | 1 & 2 | <p>Nyquist bandwidth formula (for noiseless channel, Shannon's channel capacity (for noisy channel))</p> <p><b>Analog and Digital Transmission</b></p> <p>Overview of different kinds of Conversions (like analog to digital, digital to digital, digital to analog, analog to analog)</p> <p>Digital to Digital Encoding/decoding: Line Coding (Characteristics, Unipolar, Polar, Bipolar,), Block Coding.</p> <p><b>Assignment 1, Quiz 1</b></p> |
| 5  | 1 & 2 | <p><b>Multiplexing</b> Definition and need for multiplexing, Frequency Division Multiplexing (FDM), Wavelength Division Multiplexing (WDM), Time Division Multiplexing (Synchronous, Statistical), Code Division Multiplexing (CDM)</p>                                                                                                                                                                                                            |
| 6  | 1 & 2 | <p><b>Transmission modes:</b> Serial Transmission, Parallel Transmission Circuit Switched and Packet Switched Networks</p> <p>Ethernet LAN (IEEE 802.3) standard, Overview of Wireless LAN</p>                                                                                                                                                                                                                                                     |
| 7  | 1 & 2 | <p><b>Data Link Layer</b></p> <p>Functions of Data Link Layer, Framing (byte count, character-based framing ,bit-oriented framing), Flow control (stop-and-wait, sliding window, Error detection and correction: Issues related to error detection and correction, Error detection codes (parity check, CRC), Error correction codes (Hamming code), <b>Assignment 2, Quiz 2</b></p>                                                               |
| 8  | 1 & 2 | <p><b>Data Link Layer (Cont.)</b> )MAC protocols Ingredients, Propagation Vs Transmission, Random Multiple Access Protocols (Aloha, slotted Aloha, CSMA and access modes, CSMA/CD)</p>                                                                                                                                                                                                                                                             |
| 9  | 1&2   | <p><b>Mid Term Exams:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 10 | 1 & 2 | <p><b>Network Layer</b></p> <p>Connectionless vs connection oriented service, Internet Protocol (IP), IP Addressing (Logical Addressing), Internet Assigned Number Authority (IANA), IPv4 Addresses and Hierarchy, Internet Class-based Addresses, Addressing guidelines</p>                                                                                                                                                                       |
| 11 | 1 & 2 | <p><b>Network Layer (Cont ...)</b></p> <p>Subnet and Sub-netting, Classless Addressing, VLSM, IPv4 Datagram format,</p> <p>Overview of Address Resolution Protocol (ARP), Internet Control Message Protocol (ICMP), Internet Group Message Protocol (IGMP), IPv6 Addressing , <b>Assignment 3, Quiz 3</b></p>                                                                                                                                      |
| 12 | 1 & 2 | <p><b>Network Layer (Cont ...)</b></p> <p>Packet Delivery (Direct, Indirect), Forwarding Techniques (Next hop vs route method, Network specific Vs host specific, Forwarding process, Autonomous systems.</p>                                                                                                                                                                                                                                      |
| 13 | 1 & 2 | <p><b>Network Layer (Cont ...)</b></p> <p>Distance vector routing, RIP routing protocol</p> <p>Brief overview OSPF</p> <p>BGP, <b>Assignment 4,</b></p>                                                                                                                                                                                                                                                                                            |

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| 14               | 1&2                                                                                        | <b>Transport Layer</b><br>Overview of Transmission Control Protocol, User Datagram Protocol, SCTP, Ports. <b>Quiz 4</b>                                                                                        |
| 15               | 1&2                                                                                        | <b>Application Layer Protocols</b><br>Hyper Text Transfer Protocol (HTTP and HTTPS), File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP), POP3 /IMAP4, Telnet, Domain Name Systems (DNS), DHCP. |
| 16               | 1&2                                                                                        | Course Revision                                                                                                                                                                                                |
|                  |                                                                                            | <b>Final Term Exams:</b>                                                                                                                                                                                       |
| <b>Approval:</b> |                                                                                            |                                                                                                                                                                                                                |
|                  | <b>Cluster Head / TAI:</b><br><br><i>M. Nadeem Akhtar</i><br><i>( Principal Lecturer )</i> | <b>Approved by:</b><br><br><i>Dr. Mehtab Afzal</i><br><i>(HOD CS &amp; IT)</i>                                                                                                                                 |